Machine Learning - ITA0605

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1. Perception model: Student Pass / Fail Prediction

Given

* Study Hours (x_1) = 2
* Attendance (x_2) = 1
* Weight for study Hours (W_1) = 0.6
* Weight for Attendance (W_2) = 0.4
* Bias (b) = - 0.5

Step 1: calculate Net Input:-

Formula:

Net Input = $(x_1 W_1) + (x_2 W_2) + b$

Substitute the values:

$= (2 \times 0.6) + (1 \times 0.4) + (-0.5)$

$= 1.2 + 0.4 - 0.5$

$= 1.1$

Step 2: Apply step Activation function.

Step function:

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$1.1 > 0$

Output = 1

Conclusion

Output = 1

Therefore, the student Passes.

1/8/26

2. Backpropagation Learning Rule:

Eg: updating the weight of a single Neuron

Given:-

⇒ Initial weight $(W) = 0.5$

⇒ Learning Rate $(\eta) = 0.1$

⇒ Input $(x) = 2$

⇒ Actual output $= 0.6$

⇒ Target output $= 1$

Step 1 :- Calculate Error

Formula : Error = Target − Actual

$= 1 - 0.6$

$= 0.4$

Step 2 : Calculate weight change:-

Formula :

$\Delta w = \eta \times \text{Error} \times \text{Input}$

$= 0.1 \times 0.4 \times 2$

$= 0.08$

Step 3 : update weight:

Formula :

New weight = old weight + Δw

$= 0.5 + 0.08$

$= 0.58$

Final Answer:

⇒ Error $= 0.4$

⇒ weight update $= 0.08$

⇒ New weight $= 0.58$

5. Confusion matrix:- [Disease prediction model]

Given:

TP = 180

TN = 150

FP = 20

FN = 50

Step1 : Accuracy

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{180 + 150}{180 + 150 + 20 + 50}$$

$$= \frac{330}{400}$$

$$= 0.825$$

Accuracy = 82.5%

Step 2: Precision:

Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{180}{180 + 20}$$

$$= \frac{180}{200}$$

$$= 0.9$$

Precision = 90%

Step3: Sensitivity (Recall):

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{180}{180 + 50}$$

$$= \frac{180}{230}$$

$$= 0.7826$$

Sensitivity = 78.26%

Step4: Specificity [~~Recall~~]

Formula

$$\text{sensitivity} = \frac{\cancel{TP}}{\cancel{TN} + \cancel{FP}} \Rightarrow \frac{\cancel{180}}{\cancel{180+50}} \Rightarrow \frac{\cancel{180}}{\cancel{230}}$$

$$= \cancel{0.7826} \quad \frac{150}{150+20} \Rightarrow \frac{150}{170}$$

$$= 0.8824$$

specificity = 88.24%

Conclusion

⇒ Accuracy = 82.5%

⇒ Precision = 90%

⇒ Sensitivity = 78.26%

⇒ Specificity = 88.24%.

The model Performs well overall with high Precision and specificity. However, the sensitivity is lower, meaning some diseased Patients are missed (false negatives). For medical diagnosis, improving sensitivity is important to reduce missed cases.

4.

Genetic Algorithm (maximizing the function $f(x) = x^2$)

Given Population

individual (x)	fitness $f(x) = x^2$
2	4
4	16
6	36
8	64

Step 1: Best candidates

Highest fitness values:

⇒ 8 → 64

⇒ 6 → 36

These are selected for crossover

Step 2:- Binary Representation.

6 = 0110

8 = 1000

Exchange the least significant Bit (LSB)

Before

0110

1000

After swapping LSB

0110 → 0110 (6)

1000 → 1000 (8)

offspring

6

8

Role of Mutation → Mutation randomly changes one or more bits in a Chromosome.

Ex

0110 | After mutation

0111 → 6 become 7

Importance

→ Maintains population diversity
→ Prevents premature convergence.

5. Deep Neural Network [Training and validation Accuracy Analysis]

Given

Epoch	Training Accuracy	Validation Accuracy
Beginning	60%	58%
Middle	85%	72%
~~Later~~	95%	68%

Analysis

Training accuracy continuously increases from 60% to 95%.

Validation accuracy increases initally from 58% to 72%, but later decreases to 68%.

This indicates overfitting.

Significance of validation accuracy

→ Measures the model's performance on unseen data.

→ Helps detect overfitting or underfitting

→ Assists in selecting the best model.

→ Indicates how well the model will generalize to real-world data

Conclusion

The model suffers from over fitting after several epochs.